LeaseGuard AI

1. Project Overview

LeaseGuard AI is a RAG-based AI assistant that checks key risk factors in real estate lease agreements and answers user questions based on statutes, public checklists, and curated reference documents.

The purpose of this project is to help non-legal users understand key clauses in lease agreements and organize questions that should be confirmed with a landlord or licensed real estate agent. In particular, the service provides reference-based risk checks around items such as deposit return, special clauses, repair cost responsibility, move-in registration and fixed date, registry record review, and jeonse fraud prevention checklists.

This service is not a legal advisory service. It does not determine whether a contract should be signed, whether a contract is invalid, whether a clause is illegal, or whether litigation would succeed. The answers provided by this system are reference-based risk checks grounded in contract snippets and reference sources, and expert consultation is recommended when final judgment is required.

1.1 Planning Background

Lease agreements often contain legal terms and practical expressions that are difficult for general users to understand. Deposit return timing, tenant repair cost responsibility, special clauses, move-in registration and fixed date, and senior rights shown in registry records are representative confirmation items that may lead to disputes after contract execution.

LeaseGuard AI extracts relevant sentences from an uploaded contract, searches them together with reference documents stored in ChromaDB, and generates an answer through the OpenAI Chat API. The answer includes sources so that the user can verify the supporting sentences.

1.2 MVP Scope

The current MVP operates with a non-login anonymous session model. Membership registration, JWT authentication, automated registry OCR verification, market price API integration, and automatic guarantee insurance eligibility judgment have not been implemented.

The implemented MVP scope is as follows.

- Anonymous session creation and `localStorage` storage
- TXT/PDF contract upload
- Text-extractable PDF processing
- Contract chunking and ChromaDB storage
- Curated reference document indexing
- Rule-based contract risk item analysis
- Contract source text evidence extraction
- RAG-based contract Q&A
- OpenAI Chat API-based answer generation
- Response mode-based answer strategy adjustment
- Recent conversation context delivery

- Answer source display
- Chat history and source storage
- Previous contract list retrieval, analysis revisit, and chat restoration
- Contract soft delete
- Browser anonymous session reset

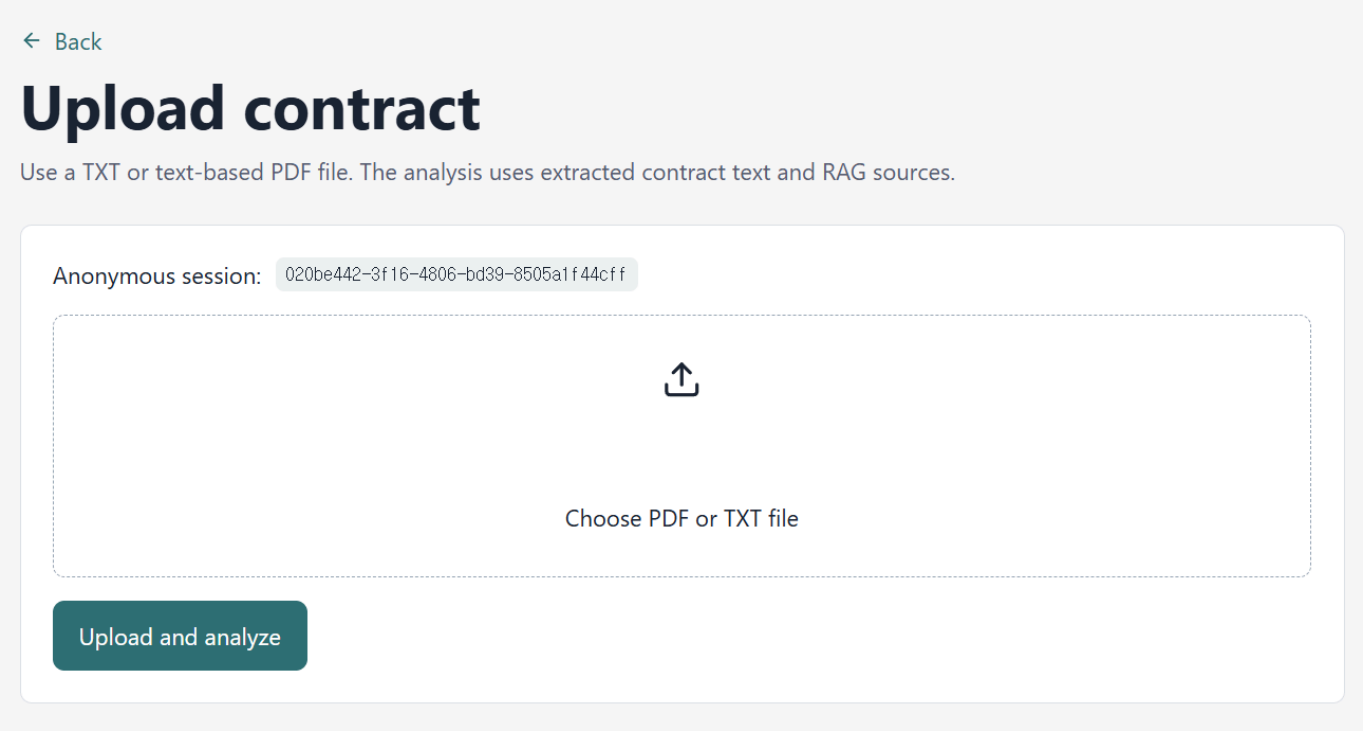
2. Key Features

Feature	Implementation
Anonymous session	Creates a UUID-based `anonymousSessionId` without login and stores it in browser `localStorage`.
Contract upload	Uploads TXT/PDF files from React through multipart form data, and Spring Boot stores the file.
PDF text extraction	FastAPI processes text-extractable PDF files with `pypdf`. Scanned PDFs return an OCR-not-supported error.
Contract indexing	Splits contract text into chunks and stores them in the ChromaDB `user_contracts` collection.
Reference indexing	Reads curated reference documents and manifest metadata, then stores them in the ChromaDB `legal_reference` collection.
Risk analysis	Performs rule-based checks for deposit return, special clauses, repair costs, contract termination, and management fees.
Evidence display	Extracts related sentences or surrounding context from the contract source text for each risk item.
RAG search	Searches contract chunks and reference chunks together and applies category-based reranking.
OpenAI answer	Generates answers through the OpenAI Chat API based on retrieved sources and chatHistory.
Response mode	Applies answer strategies such as easy explanation, analogy, landlord question phrasing, brief summary, clause rewrite example, and legal judgment refusal.
Chat restoration	Reloads stored chat sessions and messages to display previous conversations.
Dashboard	Retrieves previous contracts by anonymous session and supports analysis revisit or chat revisit.
Contract deletion	Soft-deletes contracts from the Dashboard and excludes them from the visible list.

3. Screen Examples

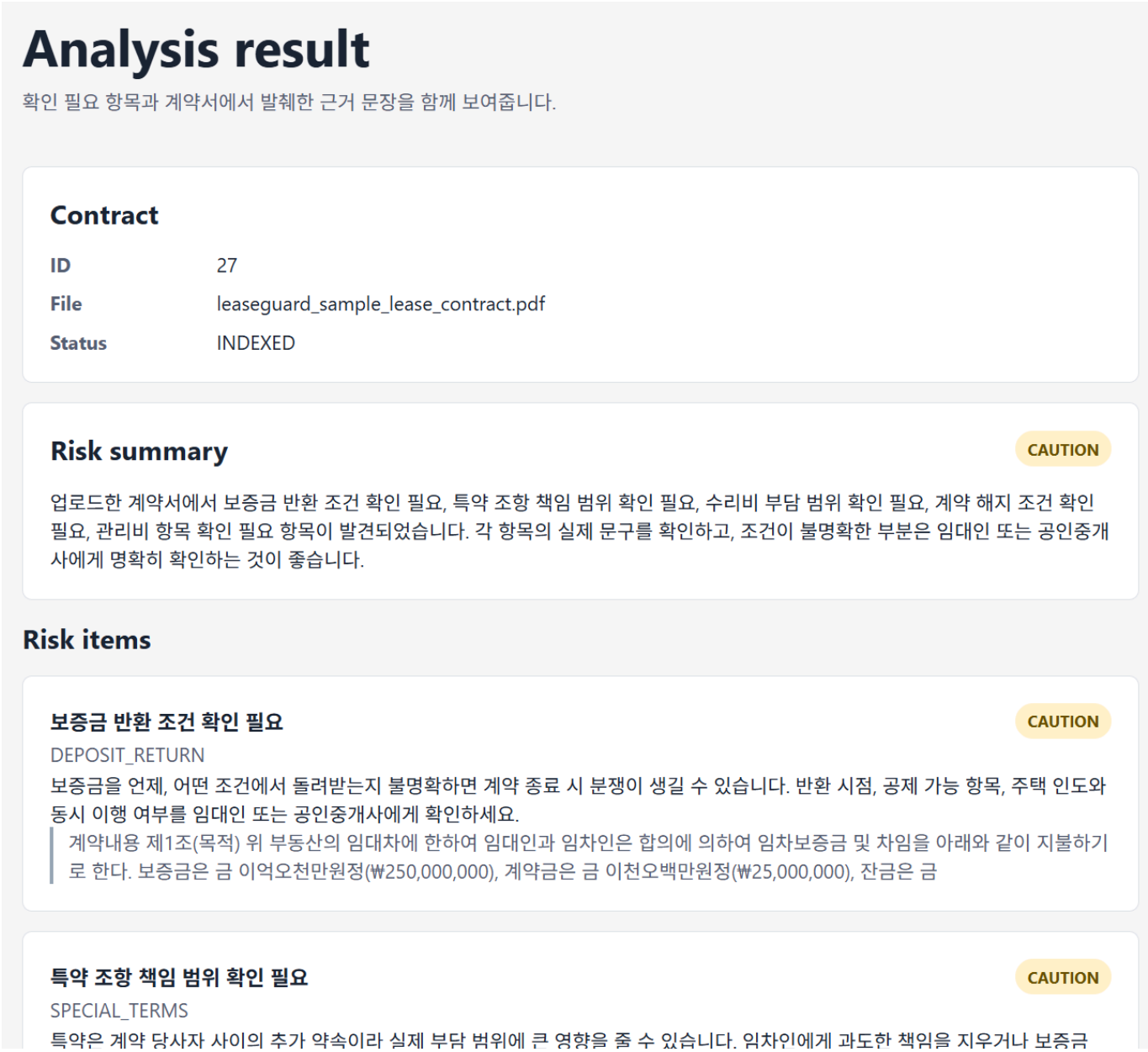
Actual screenshot files can be stored later under `docs/screenshots/`. This README keeps placeholder paths for report structure.

3.1 Contract Upload Screen



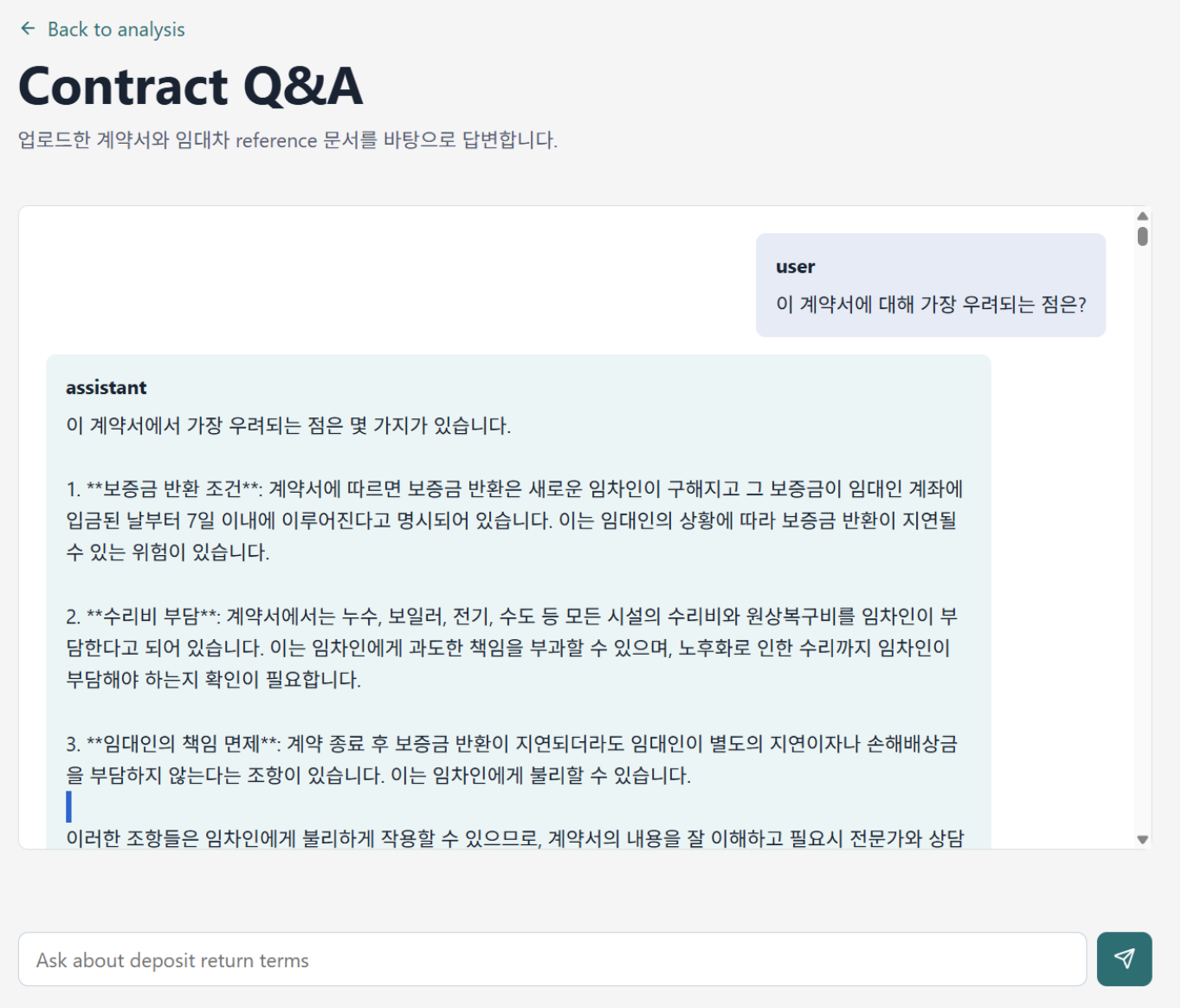
The user selects and uploads a TXT/PDF contract file. After upload, Spring Boot requests contract indexing and analysis from the FastAPI RAG server.

3.2 Analysis Result Screen



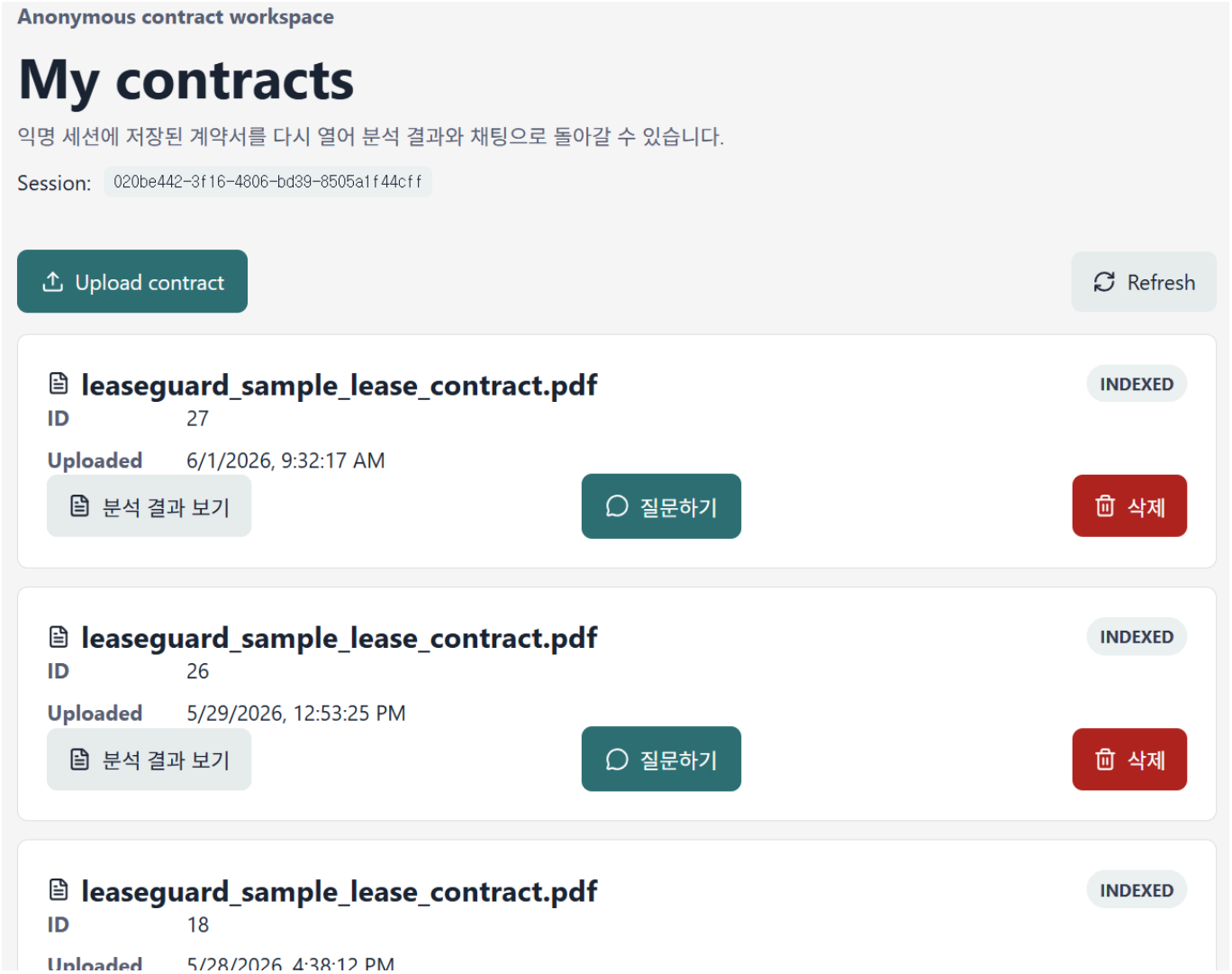
The screen displays overall risk level, summary, risk items, descriptions, and contract source text evidence. The user can move directly from the analysis result to the chat screen.

3.3 RAG Chat Screen



The screen displays user questions and assistant answers. Assistant answers preserve line breaks, and contract evidence and reference evidence are separated as sources.

3.4 Dashboard Screen



The Dashboard displays the list of uploaded contracts for the anonymous session. Each contract provides analysis result viewing, question asking, and deletion features.

3.5 Response Mode Example Screen

user

잘 모르겠는데 쉽게 설명좀

assistant

계약서의 내용을 쉽게 설명해 드릴게요.

1. **보증금 반환 조건**: 계약서에 따르면, 보증금을 돌려받으려면 새로운 임차인이 들어와야 하고, 그 임차인의 보증금이 임대인 계좌에 입금된 후 7일 이내에 반환된다고 되어 있어요. 이건 임대인이 새로운 세입자를 구하는 데 시간이 걸릴 수 있다는 점에서 위험할 수 있습니다.

2. **수리비 부담**: 계약서에서는 누수나 전기, 수도 등 모든 시설의 수리비를 임차인이 부담한다고 명시되어 있어요. 이는 임차인에게 많은 책임을 부과할 수 있으니, 어떤 경우에 수리비를 내야 하는지 잘 확인해야 해요.

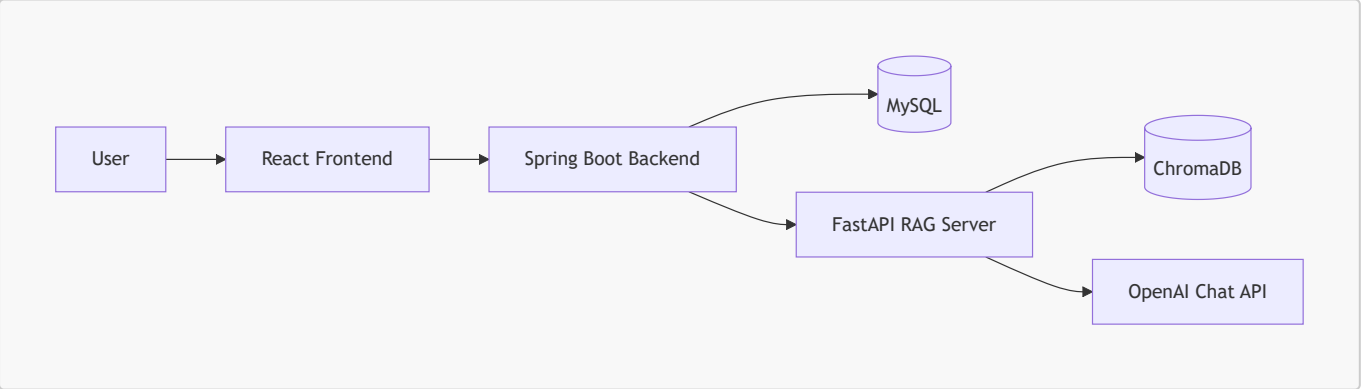
3. **임대인의 책임 면제**: 계약 종료 후 보증금 반환이 지연되더라도 임대인이 지연이자나 손해배상금을 지급하지 않겠다는 조항이 있어요. 이건 임차인에게 불리할 수 있습니다.

이런 조항들은 임차인에게 불리하게 작용할 수 있으니, 계약서를 잘 이해하고 필요하면 전문가와 상담하는 것이 좋습니다.

본 답변은 법률 자문이 아니라 참고용 위험 점검입니다.

This screen can show how answer strategies change depending on user intent, such as easy explanation, analogy, landlord question phrasing, or brief summary.

4. System Architecture



4.1 Component Roles

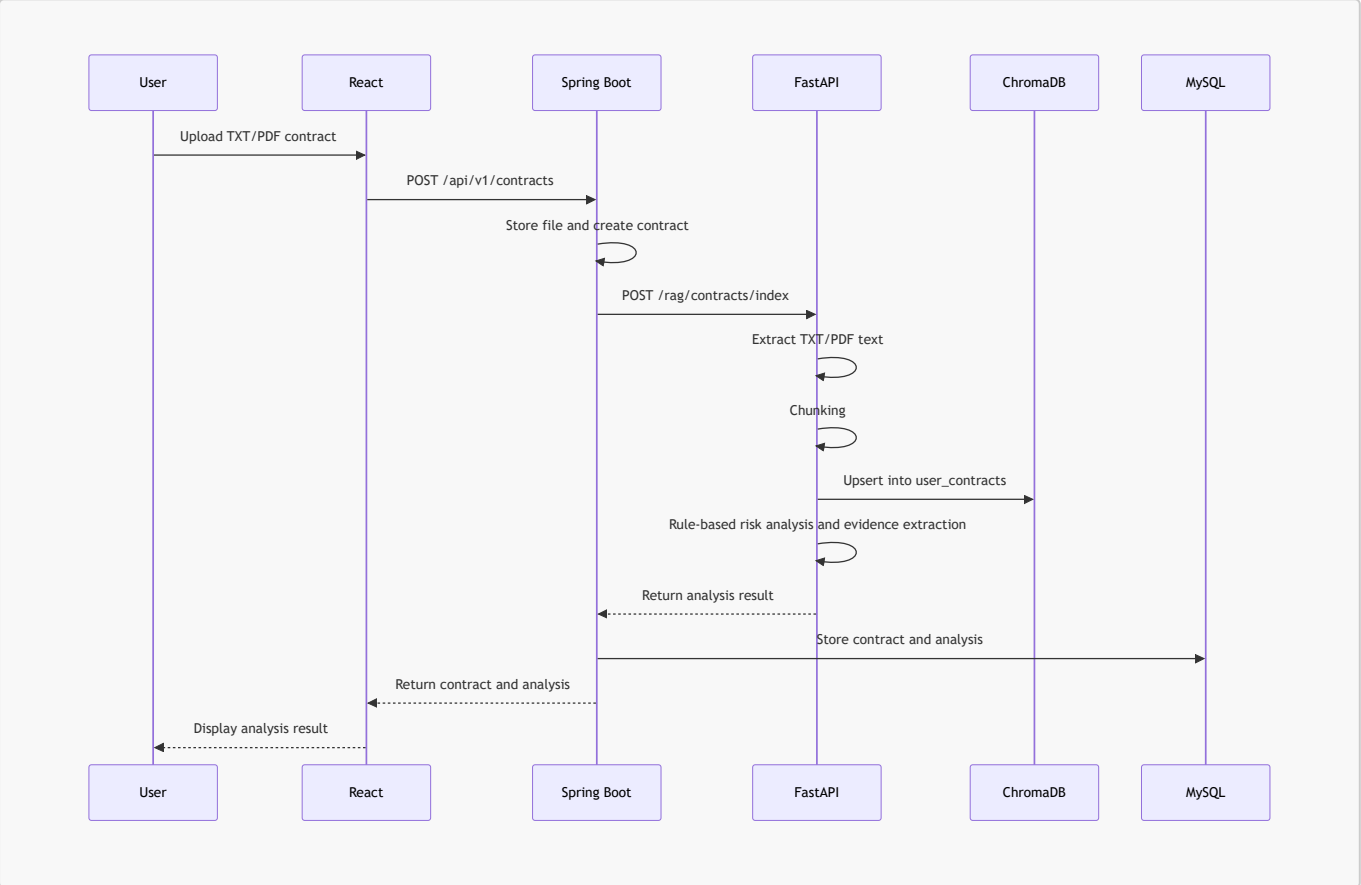
Component	Role
React Frontend	Provides contract upload, analysis result, chat, and Dashboard UI.
Spring Boot Backend	Stores anonymous sessions, contract metadata, analysis results, chat sessions, messages, and sources in MySQL.
FastAPI RAG Server	Performs contract text extraction, chunking, ChromaDB indexing, RAG search, and OpenAI calls.

Component	Role
ChromaDB	Stores and searches reference chunks and contract chunks through the `legal_reference` and `user_contracts` collections.
MySQL	Stores anonymous sessions, contracts, analysis results, chat history, and message sources.
OpenAI Chat API	Generates natural-language answers based on retrieved sources and chatHistory.

React does not call the OpenAI API or FastAPI directly. All requests follow the `React → Spring Boot → FastAPI → OpenAI API` flow.

5. Data Flow

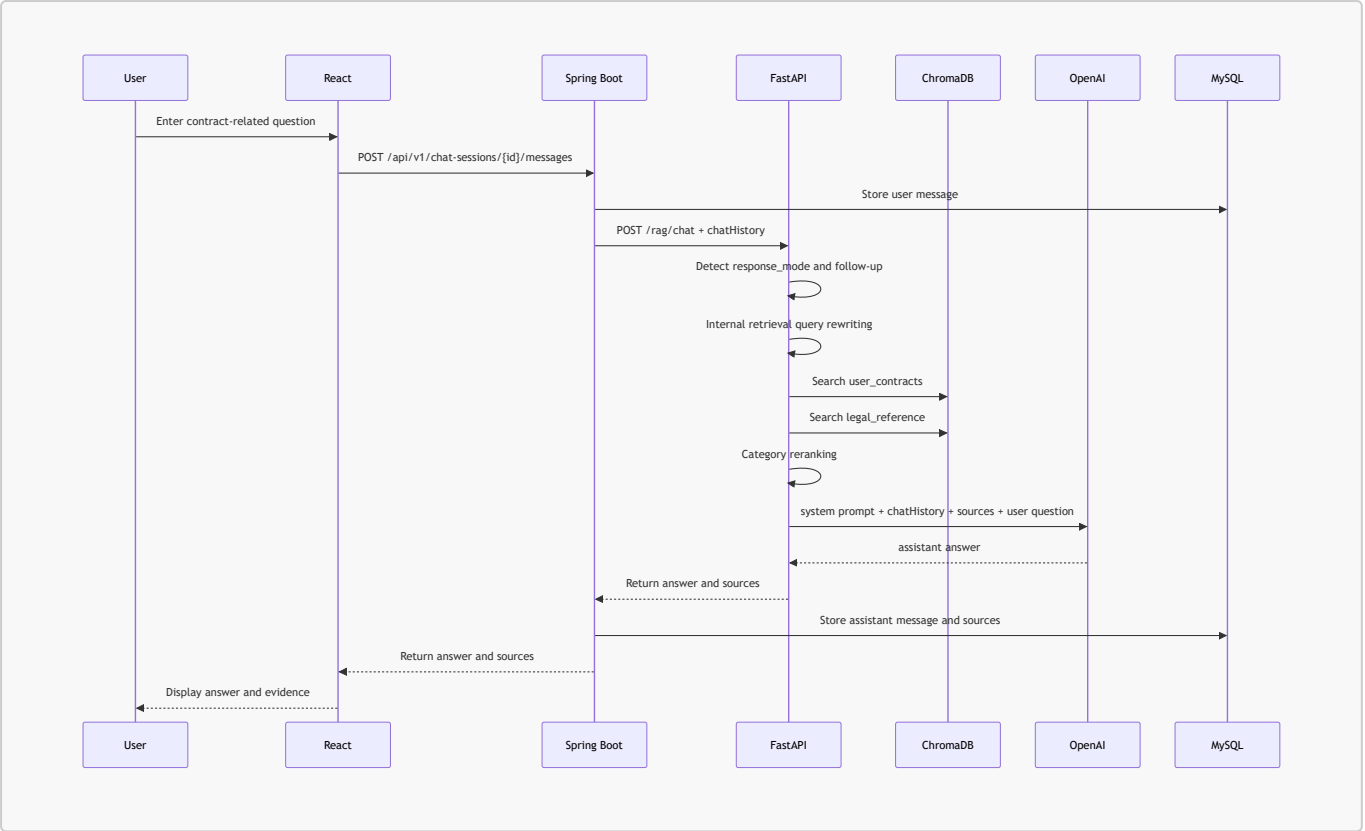
5.1 Contract Upload and Analysis Flow



5.2 Reference Document Indexing Flow

1. FastAPI reads documents from `rag-server/data/reference_sources/curated`.
2. It loads `source_manifest.json` and builds metadata such as category, sourceType, and keywords.
3. It splits documents into character-length-based chunks.
4. It creates stable ids and upserts chunks into the ChromaDB `legal_reference` collection.
5. It prevents excessive duplicate storage even when the same documents are indexed again.

5.3 Chat/RAG Answer Flow



6. AI/RAG Implementation Details

6.1 Reason for Applying RAG

This project does not directly train the model on statutes and checklists. Instead, it stores reference documents in ChromaDB and retrieves related chunks when a question is asked. This RAG approach is suitable for providing source-based explanations and controlling the model so that it does not assert content outside the provided materials.

6.2 Reference Dataset

Reference documents consist of curated txt files and a manifest.

```
rag-server/data/reference_sources/
├── curated/
└── source_manifest.json
```

The current reference documents cover the following topics.

- Deposit return
- Special clauses
- Repair cost responsibility and restoration
- Move-in registration and fixed date
- Registry records and senior rights review
- Jeonse fraud prevention checklist
- Standard contract criteria

6.3 ChromaDB Collections

Collection	Stored Data	Main Metadata
`legal_reference`	Statutes, guides, checklists, and standard contract reference documents	`category`, `sourceType`, `title`, `fileName`, `chunkIndex`, `keywords`
`user_contracts`	Contract chunks uploaded by users	`anonymousSessionId`, `contractId`, `documentName`, `chunkIndex`

Searches against `user_contracts` always use filters based on `anonymousSessionId` and `contractId`. This isolates user contract chunks so that another user's contract cannot be mixed into search results.

6.4 Search Quality Improvements

The current RAG search uses the following techniques.

- Character-length-based chunking
- Custom hash/token n-gram-based embedding
- Query expansion
- Category metadata-based reranking
- Contract/reference source mix preservation
- Search quality checks based on test questions

OpenAI embedding is not currently used. LangChain is also not used in the current MVP.

6.5 OpenAI Answer Generation

FastAPI builds sources from ChromaDB search results and passes them to the OpenAI Chat API. The default model is `gpt-4o-mini`, and it can be changed through the `OPENAI_MODEL` environment variable.

OpenAI call conditions are as follows.

- If `OPENAI_API_KEY` exists, the OpenAI Chat API is called.
- If the API key is missing or the call fails, a template fallback answer is returned.
- The fallback answer also preserves the message that the response is a reference-based risk check, not legal advice.
- If no sources exist, the answer states that it is difficult to confirm based only on the provided materials.

6.6 LeaseGuard AI Persona

LeaseGuard AI does not make final judgments like a legal expert. It explains difficult clauses in plain language and helps users organize questions to ask a landlord or licensed real estate agent.

The answer principles are as follows.

- Answers are grounded in contract snippets and reference sources.
- Content not present in sources is not asserted.
- Contract invalidity, illegality, litigation outcome, or contract signing possibility is not determined.
- Expert consultation is recommended when necessary.

- The answer includes the meaning that it is a reference-based risk check, not legal advice.

6.7 Response Mode-Based Prompt Routing

`response_mode` is not a fixed output template. It is used as a hint that determines the answer strategy and tone.

response_mode	Purpose
`structured_analysis`	Default risk analysis answer
`easy_explanation`	Easy explanation
`analogy`	Analogy or example-centered explanation
`landlord_question`	Suggested sentences to ask the landlord
`brief_summary`	Short key summary
`rewrite_clause`	Clause revision direction and reference example wording
`legal_judgment_refusal`	Avoidance of final legal judgment and guidance on confirmation items

6.8 Conversation Context Maintenance

Spring Boot passes recent messages to FastAPI as `chatHistory`. FastAPI includes them in OpenAI messages so that follow-up questions such as “that part,” “the clause mentioned earlier,” or “what should I do then” can be interpreted according to the previous conversation flow.

FastAPI also performs internal retrieval query rewriting. This process does not change the original user question shown to the user, and it is used only internally for ChromaDB search and OpenAI prompt construction.

7. Backend Implementation Details

The Backend is a Spring Boot-based API server. It acts as the middle layer between React and FastAPI, and it handles data persistence and user request validation.

7.1 Package Roles

Package	Role
`anonymous`	Anonymous session creation and retrieval
`contract`	Contract upload, list retrieval, detail retrieval, analysis result retrieval, and soft delete
`chat`	Chat session creation, message storage, source storage, and chatHistory delivery
`rag`	FastAPI RAG server client and DTOs
`global`	Common response, exception handling, and configuration

7.2 Main APIs

Method	Endpoint	Description
POST	`/api/v1/anonymous-sessions`	Create anonymous session
POST	`/api/v1/contracts`	Upload and analyze contract
GET	`/api/v1/contracts`	Retrieve contract list
GET	`/api/v1/contracts/{contractId}`	Retrieve contract details
GET	`/api/v1/contracts/{contractId}/analysis`	Retrieve analysis result
DELETE	`/api/v1/contracts/{contractId}`	Soft-delete contract
POST	`/api/v1/chat-sessions`	Create chat session
GET	`/api/v1/chat-sessions`	Retrieve chat session list
GET	`/api/v1/chat-sessions/{chatSessionId}/messages`	Retrieve message list
POST	`/api/v1/chat-sessions/{chatSessionId}/messages`	Send message and generate RAG answer

8. FastAPI RAG Server Implementation Details

FastAPI handles contract analysis and RAG search.

Method	Endpoint	Description
GET	`/health`	Check RAG server status
POST	`/rag/references/index`	Index reference documents
POST	`/rag/contracts/index`	Index contract and perform rule-based analysis
POST	`/rag/chat`	Perform RAG search and generate OpenAI answer

8.1 Main Service Files

File	Role
`contract_parser.py`	TXT/PDF text extraction
`chunking_service.py`	Document chunking
`reference_indexing_service.py`	Reference document indexing
`contract_indexing_service.py`	Contract indexing
`retrieval_service.py`	ChromaDB search, query expansion, and reranking
`llm_service.py`	OpenAI call, response mode, fallback, and prompt construction
`risk_analysis_service.py`	Rule-based risk item analysis and evidence extraction
`chroma_client.py`	ChromaDB client and collection management

9. Frontend Implementation Details

The Frontend is implemented with React and Vite. It uses minimal styling while keeping the interface suitable for feature verification.

Screen	Feature
Home	Service introduction and upload start
ContractUpload	TXT/PDF contract upload and analysis request
Analysis	Risk level, summary, risk items, and evidence display
Chat	Chat messages, assistant answer, and sources display
Dashboard	Previous contract list, analysis revisit, chat revisit, deletion, and session reset

The Frontend stores `anonymousSessionId` in `localStorage` and includes the `X-Anonymous-Session-Id` header in all API requests.

10. Execution Guide

10.1 Prerequisites

- Java 17
- Node.js and npm
- Python 3.x
- Docker Desktop
- OpenAI API key

10.2 Environment Variables

Configure `.env` based on `.env.example` in the project root.

```
# MySQL
MYSQL_ROOT_PASSWORD=root
MYSQL_DATABASE=leaseguard
MYSQL_USER=leaseguard
MYSQL_PASSWORD=leaseguard123

# Spring Boot
SPRING_DATASOURCE_URL=jdbc:mysql://localhost:3306/leaseguard
SPRING_DATASOURCE_USERNAME=leaseguard
SPRING_DATASOURCE_PASSWORD=leaseguard123
RAG_SERVER_BASE_URL=http://localhost:8000

# FastAPI
OPENAI_API_KEY=your_openai_api_key
OPENAI_MODEL=gpt-4o-mini
CHROMA_HOST=localhost
CHROMA_PORT=8001
```

The actual `.env` file is not included in Git. The OpenAI API key is not exposed to the frontend.

10.3 Run MySQL and ChromaDB

```
docker compose up -d mysql chromadb
```

MySQL uses host `localhost` and port `3306`. ChromaDB uses host `localhost` and port `8001`.

10.4 Run Backend

```
cd backend
.\gradlew.bat bootRun
```

The default Backend port is `8080`.

10.5 Run FastAPI

```
cd rag-server
.\.venv\Scripts\Activate.ps1
pip install -r requirements.txt
uvicorn app.main:app --reload --port 8000
```

The default FastAPI port is `8000`.

10.6 Run Frontend

```
cd frontend
npm.cmd install
npm.cmd run dev
```

The default Frontend port is `5173`. If PowerShell has execution issues, use `npm.cmd` instead of `npm`.

11. Test Scenarios

11.1 Contract Upload and Analysis

1. Create an anonymous session.
2. Upload a TXT contract or a text-extractable PDF contract.
3. Spring Boot calls FastAPI `/rag/contracts/index`.
4. FastAPI extracts contract text and stores it in ChromaDB.
5. FastAPI returns rule-based analysis results and evidence.
6. React displays risk items on the analysis result screen.

11.2 RAG Chat

Example questions are as follows.

- What is the riskiest part of this contract?
- Check whether the deposit return condition is risky.
- Are there any special clauses that are unfavorable to the tenant?
- Why are move-in registration and fixed date necessary?
- What should be checked in the registry record?

11.3 Follow-up Questions

An example question flow is as follows.

1. What is the riskiest part of this contract?
2. Then what can I realistically do?
3. How should I ask the landlord about the clause mentioned earlier?
4. How should that clause be revised?

11.4 Response Mode

Example questions are as follows.

- This is too difficult. Explain it simply.
- Explain it using an analogy.
- What should I ask the landlord?
- Tell me only the key points briefly.
- How should I revise this clause?
- Is this contract invalid? Would I win if I sued?

12. Verification Results

The following items were verified during development.

Verification Item	**Result**
Spring Boot compile	Success
React build	Success
FastAPI compileall	Success
FastAPI app import	Success
`/rag/references/index`	Success
`/rag/contracts/index`	Success
`/rag/chat` fallback	Success
`/rag/chat` OpenAI call	Success
Contract excluded from list after deletion	Success

Verification Item	Result
Deleted contract analysis and chat access blocked	Success
New anonymous session created after session reset	Success

Detailed API test commands are documented in `TEST_COMMANDS.md` and `TEST_RAG_SEARCH.md`.

13. Troubleshooting

Issue	Cause and Resolution
`curl` behaves unexpectedly in PowerShell	Use `curl.exe` or `Invoke-RestMethod` because of PowerShell alias behavior.
Korean text appears as `?` in PowerShell 5	This is a console input/output encoding issue. React and API JSON are processed normally with UTF-8.
`vite` command not found	Run `npm.cmd install` in the `frontend` directory first.
npm execution policy issue	Use `npm.cmd` in PowerShell.
port `8080` already in use	Terminate the existing Spring Boot process or change the port.
ChromaDB telemetry warning	This warning does not affect functional behavior.
`message_sources.source_type` length exceeded	The DB column length was expanded and FastAPI `sourceType` was normalized.
Scanned PDF upload failure	OCR is not implemented, so only text-extractable PDFs are supported.
OpenAI API call failure	Check `OPENAI_API_KEY`, billing status, and network availability.

14. Current Limitations

- Scanned PDF and image OCR are not supported.
- OpenAI embedding is not used.
- LangChain and LangSmith are not currently used.
- The reference document scope is centered on the MVP validation curated dataset.
- Legal judgment is not finalized.
- Anonymous sessions are based on browser `localStorage`, so accessing previous lists becomes difficult when the browser changes.
- Actual service deployment, authentication, rate limiting, and security hardening have not been implemented.
- Automated registry verification, market price comparison, and guarantee insurance eligibility judgment have not been implemented.

15. Future Development Directions

- OCR-based processing for scanned PDF and image contracts

- Review of OpenAI embedding or Korean sentence-transformers-based semantic embedding
- Expansion of statute article-level references
- Improvement of source citation UI
- Automatic clauseType classification by contract clause
- Advanced risk scoring
- User account-based document management
- Deployment environment configuration and security hardening
- Registry record OCR analysis
- Address-based market price and transaction price comparison
- Connection to lawyer or licensed real estate agent consultation

16. Retrospective

This project confirmed that RAG search quality and metadata design affect answer quality more directly than a simple LLM call. How reference documents are chunked and what categories and keywords are assigned significantly influences search results and source quality.

In the legal domain, safe persona design and source-based explanations are more important than assertive answers. LeaseGuard AI was designed not to provide final legal judgment, but to help users organize risk factors and questions that should be checked in a contract.

Separating Spring Boot and FastAPI was effective for separating general service logic from AI/RAG logic. In React, implementing upload, analysis, chat, and Dashboard flows also confirmed that UX features such as conversation restoration, revisiting previous contracts, deletion, and session reset are important for actual usability.